

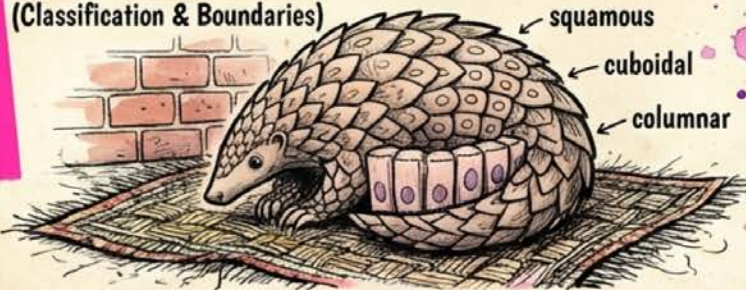
Uncle FrizzleBob's Krazy Kayfabizzaro Komik Kard Kult Klub!



Epi the Epithelium

(Classification & Boundaries)

STAT: Type IV Collagen



POWER: Barrier Seal. Blocks all non-lymphocyte tokens from crossing the board.
LORE: Basement membrane is Type IV collagen + laminin. Split it like a cheap rug, and you get Alport syndrome.

Clara the Ciliated

(Pseudostratified Columnar)

STAT: 1,000 beats/min

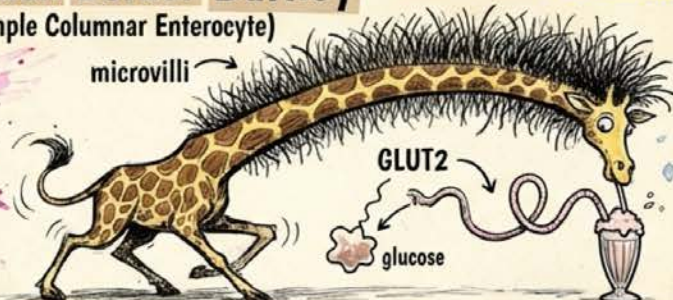


POWER: Mucociliary Escalator. Move 1 pathogen token to the Pharynx discard pile every turn.
LORE: Kartagener's syndrome = defective dynein arms. Immotile cilia mean chronic sinusitis and 50% chance of situs inversus.

Brush Border Barney

(Simple Columnar Enterocyte)

STAT: 600x Surface Area



POWER: Hyper-Absorb. Tap to digest and draw 3 nutrient cards.
LORE: Gluten wrecks Barney's hair (Villous atrophy = Coeliac disease). No hair,

Squamous Sue

(Stratified Squamous)

STAT: 15-20 Dead Layers



POWER: Keratin Armor. Sacrifice top layer to negate mechanical or chemical damage.
LORE: Pemphigus vulgaris attacks the spines (desmoglein-3) in the stratum spinosum, causing suprabasal blisters.

Urothelial Ulrich

(Transitional Epithelium)

STAT: 3 to 7 Cells Thick



POWER: Shape-Shift. Change form to nullify chemical burns from urine tokens.
LORE: Umbrella cells use uroplakin to stop urea back-diffusion. Visible haematuria? Look out for urothelial carcinoma.

Goblet the Glycoprotein

(Goblet Cells)

STAT: MUC2 (Intestine)



POWER: Mucin Blast. Create a 2-layer protective gel field.
LORE: Goblet cells in the esophagus? That's Barrett's intestinal metaplasia. Risk of adenocarcinoma goes up.

Gland Greta

(Exocrine Glands)

STAT: Amylase & Lipase



POWER: Exocrine Eruption. Deal heavy enzymatic damage to food tokens.
LORE: Parotid is purely serous (dark). Sublingual is predominantly mucous (pale). Sjogren's kills the acini.

KayfaBONGO!

Type II Pneumocyte Pete

(Alveolar Epithelium)

STAT: 5% Surface, 60% Cells



POWER: Surfactant Synthesis. Reduce alveolar surface tension to 0. Prevent collapse.
LORE: Premies lack Pete's surfactant, causing Respiratory Distress Syndrome. Antenatal steroids speed up Pete's maturation.

MESOTHELIAL MIA (Mesothelium)

STAT: WTI, Calretinin (+)



POWER: Frictionless Glide. Nullify organ friction.
LORE: Asbestos fibers pierce her, causing 20-50 year latency Mesothelioma.

COLLAGEN CARL (Collagen)

STAT: 30% Total Protein



POWER: Tensile Strength. Grant indestructible structural scaffolding.
LORE: Glycine is every 3rd position. No Vitamin C = poor prolyl hydroxylation = Scurvy (unstable collagen, bleeding).

ENDOTHELIAL EDDIE (Vascular Endothelium)

STAT: Weibel-Palade Bodies



POWER: Gatekeeper. Control leukocyte rolling and vascular tone (NO).
LORE: Liver sinusoids lack a basement membrane and have large fenestrae for free exchange.

FIBRO THE FIBROBLAST (Fibroblasts & ECM)

STAT: Alpha-SMA (+)



POWER: Matrix Remodel. Transform into Myofibroblast to contract wounds.
LORE: Alpha-1 antitrypsin deficiency allows neutrophil elastase to eat Fibro's elastin, causing emphysema.

Macro the Macrophage

(Tissue-Resident Macrophage)

STAT: CD14/CD68 (+)



POWER: Tissue Adaptation. Gain specific abilities based on organ zone.
LORE: Heart failure causes pulmonary congestion; alveolar macrophages eat leaked RBCs to become haemosiderin-laden heart failure cells.

Adipose Ada

(Adipose Tissue)

STAT: Leptin/UCP1



POWER: Thermogenesis / Satiety. Generate heat or suppress appetite.
LORE: Brown fat uses UCP1 to uncouple electron transport, making heat instead of ATP.

Mast Cell Mona

(Mast Cells)

STAT: Tryptase >20 µg/L



POWER: Degranulation. Instantly inflict massive vasodilation and bronchospasm.

LORE: Leukotrienes cause more sustained asthma bronchospasm than histamine.

Cartilage Carla

(Cartilage)

STAT: Avascular (Diffusion only)



POWER: Compressive Resistance. Absorb massive mechanical loads.
LORE: Chondrocytes have NO blood vessels; they heal terribly. Loss of Carla in joints = Osteoarthritis eburation.



Bone Barry

(Bone Microanatomy)

STAT: Hydroxyapatite (70%)



POWER: Remodel. Couple osteoclast destruction with osteoblast creation.
LORE: Paget's disease shows chaotic mosaic cement lines and giant abnormal osteoclasts.

Sarco the Sarcomere

(Skeletal Muscle)

STAT: ATP Hydrolysis



POWER: Cross-Bridge Cycle. Consume 1 ATP to deal massive mechanical force.
LORE: No ATP after death = myosin can't detach from actin = Rigor mortis.

Elastic Ellie

(Elastic Fibres)

STAT: Desmosine Crosslinks



POWER: Recoil. Return stretched tissue to resting state instantly.
LORE: Marfan syndrome = *FBN1* (fibrillin) mutation. Poor scaffold leads to aortic root dissection and lens subluxation.

Cardiac Carla

(Cardiac Muscle)

STAT: 25-35% Mitochondria



POWER: Functional Syncytium. Share electrical action potentials instantly.
LORE: Connexin-43 gap junctions allow the heart to beat as one. Ischemia breaks them, causing arrhythmias.

Smooth Sophia

(Smooth Muscle)

STAT:
Latch
State



POWER: Sustained Tone. Maintain contraction using near-zero energy.

LORE: Hirschsprung disease = absent ganglion cells in distal colon, meaning Sophia can't relax. Result: Megacolon.

Oligodendro the Myelinator

(Oligodendrocytes)

STAT:
1 cell :
40 axons

KayfaBINGO!



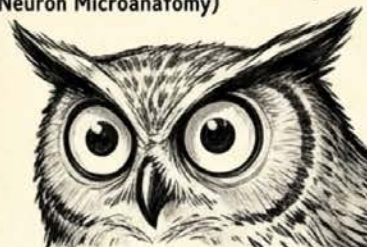
POWER: Saltatory Conduction. Increase nerve conduction velocity by 100x.

LORE: Multiple Sclerosis is a T/B-cell attack on CNS myelin, slowing conduction and leaving glial scars.

Neuron Ned

(Neuron Microanatomy)

STAT:
400 mm/day
Transport



POWER: Action Potential. Transmit signals with absolute spatial precision.

LORE: Tetanus and Rabies use retrograde transport (dynein) to hijack the axon and ride back to the CNS.

Astro the Astrocyte

(Astrocytes)

STAT:
100,000
Synapse
Contacts



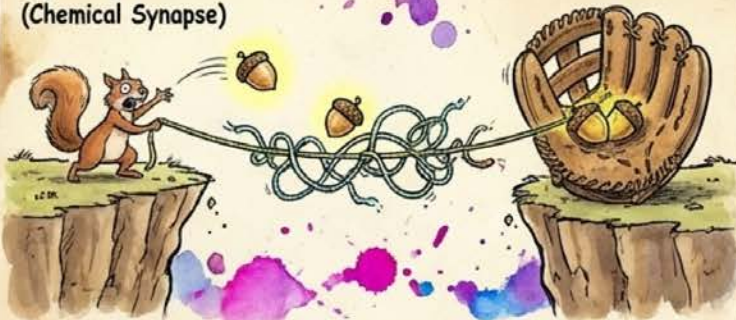
POWER: Tripartite Support. Maintain the BBB and clear neurotransmitter waste.

LORE: Hepatic encephalopathy causes Astro to swell with glutamine, raising intracranial pressure.

Synapse Sid

(Chemical Synapse)

STAT: 20-40nm Cleft



POWER: Neurotransmission. Convert electrical signal to chemical payload.
LORE: Botulinum toxin cleaves SNARE proteins. No zipline = no ACh release = flaccid paralysis.

Hepatic Lobule Hank

(Liver Microanatomy)

STAT: CYP450 Zone 3

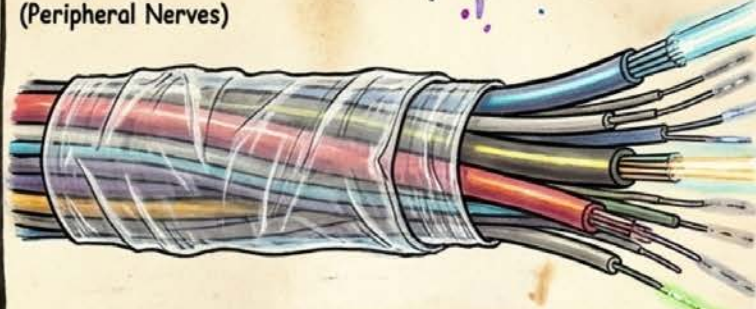


POWER: First-Pass Metabolism. Filter and detoxify all portal blood.
LORE: Zone 3 (centrilobular) has the lowest O_2 and highest CYP450. Paracetamol overdose and ischemia kill Zone 3 first.

Peripheral Nerve Percy

(Peripheral Nerves)

STAT: A δ vs C Fibers



POWER: The Blood-Nerve Barrier. Perineurium seals the microenvironment.
LORE: Local anesthetics block the thin/myelinated A δ (sharp pain) before the unmyelinated C fibers (dull pain).

Renal Corpuscle Rita

(Glomerular Microanatomy)

STAT: 180 Liters/Day



POWER: Ultra-Filtration. Separate plasma from blood cells and large proteins.
LORE: Podocyte foot process effacement (visible only on EM) destroys the barrier, causing massive proteinuria (Minimal Change Disease).

Spleen Stella

(Splenic Microanatomy)

STAT: 30% Platelet Pool



POWER: Senescent Clearance. Destroy aged/rigid RBCs and mount T-independent responses.
LORE: Asplenic patients cannot remove Howell-Jolly bodies from RBCs and are vulnerable to encapsulated bacteria.

Lymph Node Lenny

(Lymph Node)

STAT: BCL2 (Follicles)



POWER: Affinity Maturation. Evolve high-affinity antibodies over months.
LORE: Follicular lymphoma overexpresses BCL2, making B-cells immortal even when they fail the arena test.

Thymus Thomas

(Thymic Microanatomy)

STAT: >95% Apoptosis Rate



POWER: Central Tolerance. Prevent autoimmune disease via negative selection.
LORE: AIRE deficiency causes autoimmune polyendocrinopathy (APS-1). Autoreactive T-cells graduate that shouldn't.

Bone Marrow Betty

(Bone Marrow)

STAT: 200 Billion RBCs/day



POWER: Haematopoiesis. Spawn infinite blood cell tokens.
LORE: Follicular lymphoma infiltrates the marrow in a paratrabeular pattern, clinging to the bone edges.

BLÖDSINN!



Pancreatic Island Izzy

(Islets of Langerhans)

STAT: 2%
Pancreatic Volume



POWER: Endocrine Counter-Regulation. Balance blood glucose via portal paracrine signaling.
LORE: Type 1 Diabetes is a targeted T-cell strike on the central Beta bears; the Alpha wolves are spared but lose regulation.

Glomerular GBM Greta

(Glomerular Basement Membrane)

STAT: $\alpha3\alpha4\alpha5$
Collagen IV

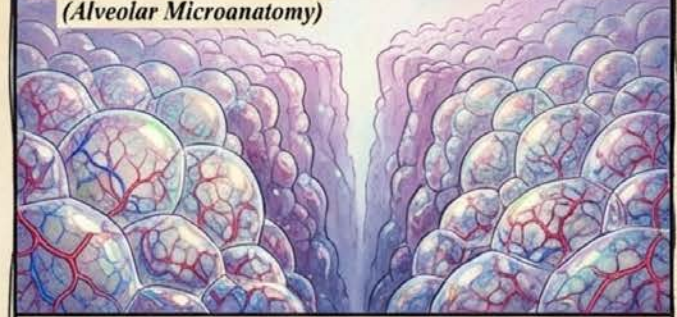


POWER: Charge & Size Barrier. Repel anionic albumin.
LORE: Alport syndrome = failed switch to $\alpha3\alpha4\alpha5$. The net splinters into a basket-weave, causing nephritis and deafness.

Alveoli-Alice

(Alveolar Microanatomy)

STAT: 0.2 μ m Barrier



POWER: Gas Exchange. Swap O₂ and CO₂ across the thinnest barrier in biology.
LORE: Fibroblastic foci and honeycombing (destroyed bubbles) are the hallmark of Usual Interstitial Pneumonia (IPF).

Gut-Wall-Ginny

(GI Wall Microanatomy)

STAT: 4 Distinct Layers



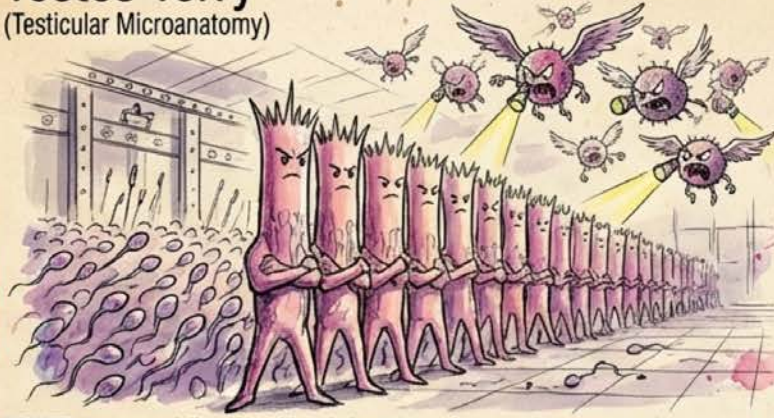
POWER: Peristaltic Digestion. Move and absorb nutrients sequentially.
LORE: Destruction of gastric parietal cells (fried-egg appearance) halts Intrinsic Factor, leading to Pernicious Anaemia.



Testes Terry

(Testicular Microanatomy)

STAT: 74-Day Cycle



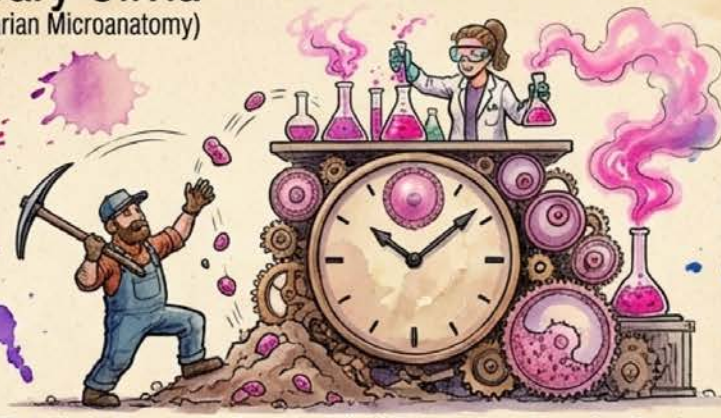
POWER: Immunoprivilege. Shield meiotic antigens from autoimmune attack.

LORE: Breach the Sertoli tight junctions, and the immune system will attack the sperm (autoimmune orchitis).

Ovary Olivia

(Ovarian Microanatomy)

STAT: Two-Cell, Two-Gonadotrophin



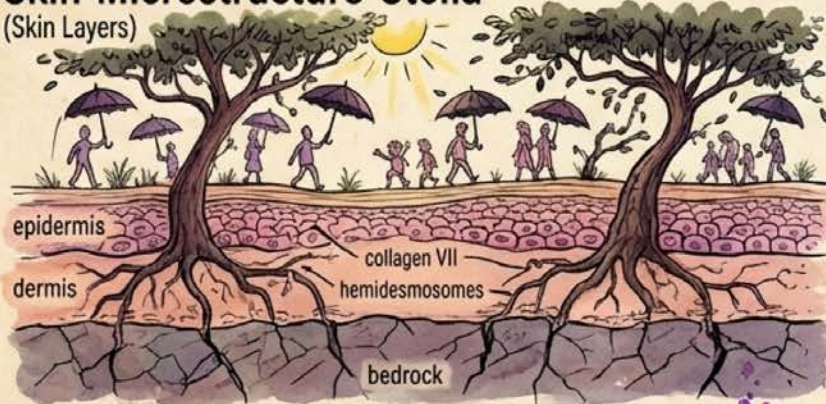
POWER: Follicular Recruitment. Burn primordial follicles to generate one Graafian champion.

LORE: PCOS: Follicles arrest in development due to excess androgens and no FSH surge, causing polycystic morphology.

Skin Microstructure Stella

(Skin Layers)

STAT: BP180 Hemidesmosomes



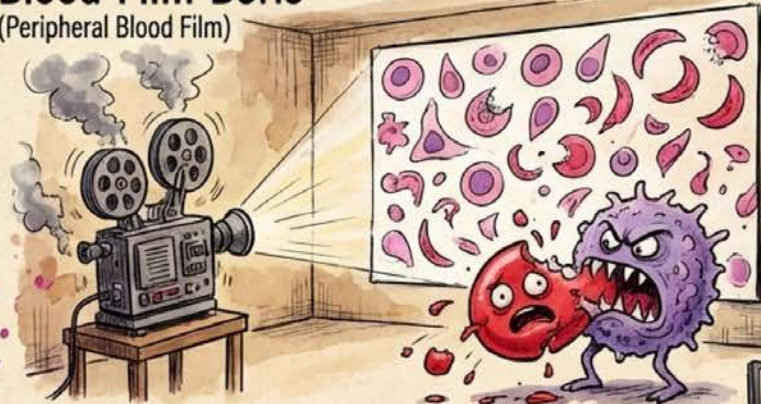
POWER: Stratified Defense. Absorb UV radiation and mechanical shear.

LORE: Bullous pemphigoid = antibodies against BP180. The epidermis cleanly detaches from the dermis (subepidermal blister).

Blood Film Boris

(Peripheral Blood Film)

STAT: The Ultimate Smear



POWER: Diagnostic Snapshot. Reveal the entire haematological state in one look.

LORE: Schistocytes + thrombocytopenia = Microangiopathic Haemolytic Anaemia (TTP/HUS/DIC). A medical emergency.

Congo Red Connie

(Amyloid)

STAT: Crossed Beta-Sheet



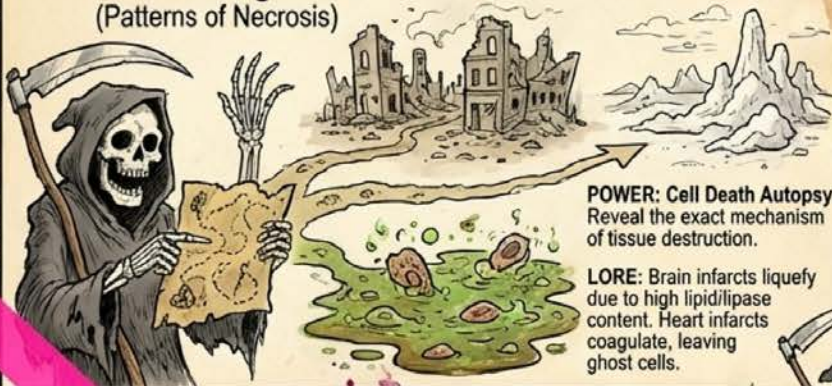
POWER: Fibrillary Accumulation.
Slowly crush organs with indestructible protein.

LORE: AL amyloid = light chains (myeloma).
AA amyloid = acute phase SAA
(chronic inflammation like RA/IBD).

Necrosis Navigator Ned

(Patterns of Necrosis)

STAT: Karyolysis, Karyorrhexis, Pyknosis



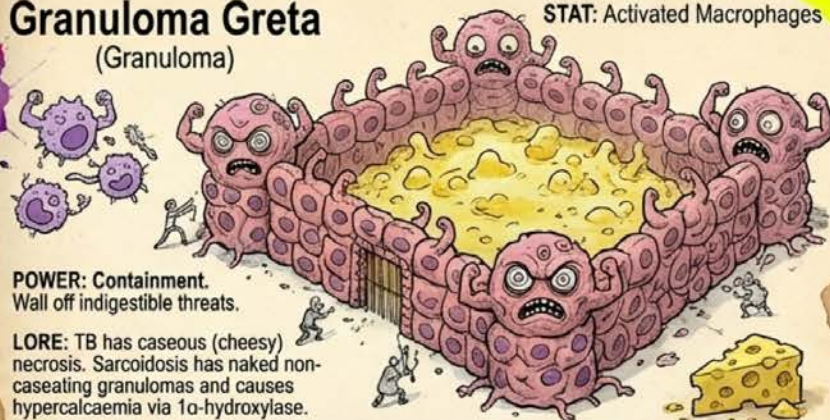
POWER: Cell Death Autopsy.
Reveal the exact mechanism of tissue destruction.

LORE: Brain infarcts liquefy due to high lipidlipase content. Heart infarcts coagulate, leaving ghost cells.

Granuloma Greta

(Granuloma)

STAT: Activated Macrophages



POWER: Containment.
Wall off indigestible threats.

LORE: TB has caseous (cheesy) necrosis. Sarcoidosis has naked non-caseating granulomas and causes hypercalcaemia via 1 α -hydroxylase.

Stain Stanley

(Special Stains)

STAT: The Decoder Toolkit



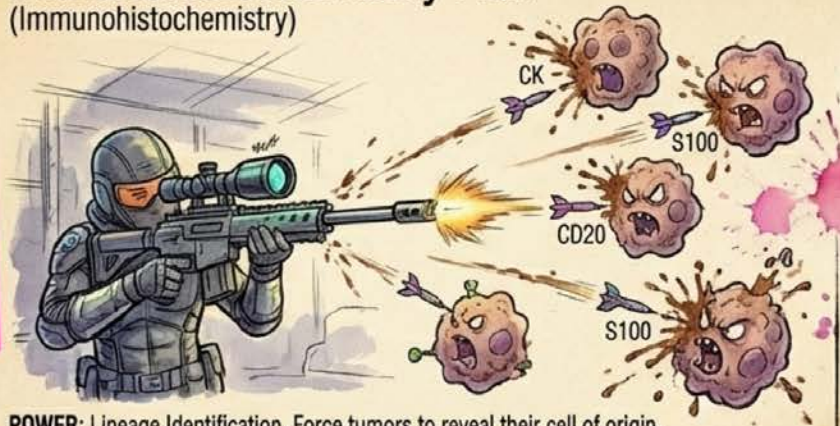
POWER: Chemical Revelation.
Reveal hidden tissue components.

LORE: H&E can't see everything. Want to see if that liver is cirrhotic? Masson's trichrome will paint the fibrotic collagen bright blue.

What the FLUFFY?

Immunohistochemistry Ivan

STAT: Brown DAB Precipitate
(Immunohistochemistry)

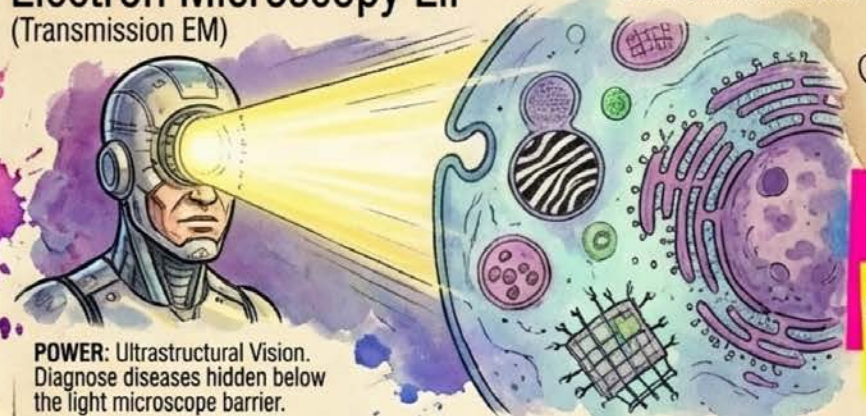


POWER: Lineage Identification. Force tumors to reveal their cell of origin.
LORE: Carcinoma = Cytokeratin+. Melanoma = S100+. Lymphoma = CD45+.

Electron Microscopy Eli

(Transmission EM)

STAT: 0.1nm Resolution



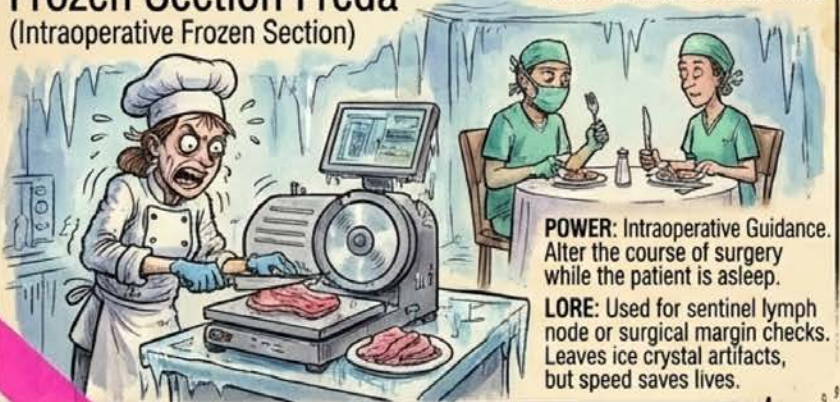
POWER: Ultrastructural Vision. Diagnose diseases hidden below the light microscope barrier.

LORE: Fabry disease shows lamellated zebra bodies. Membranous nephropathy shows subepithelial spike and dome deposits.

Frozen Section Freda

(Intraoperative Frozen Section)

STAT: 20-Minute Turnaround



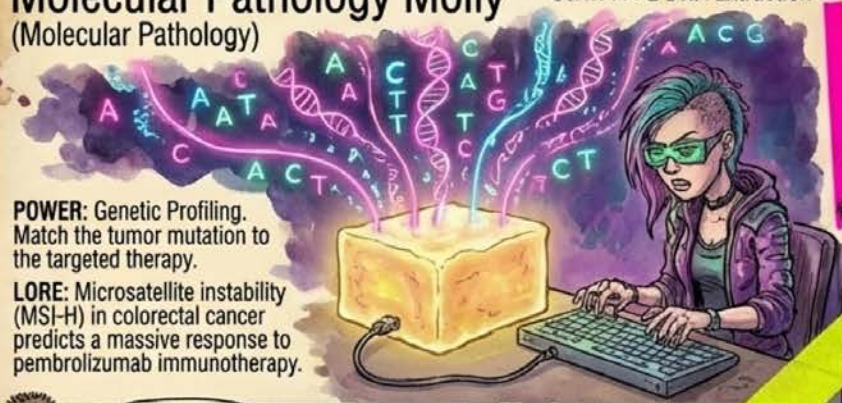
POWER: Intraoperative Guidance. Alter the course of surgery while the patient is asleep.

LORE: Used for sentinel lymph node or surgical margin checks. Leaves ice crystal artifacts, but speed saves lives.

Molecular Pathology Molly

(Molecular Pathology)

STAT: FFPE DNA Extraction



POWER: Genetic Profiling. Match the tumor mutation to the targeted therapy.

LORE: Microsatellite instability (MSI-H) in colorectal cancer predicts a massive response to pembrolizumab immunotherapy.

Kayfa8IZARRO!

Apoptosis Andy

(Apoptosis)

STAT: Caspase-3 Executioner



POWER: Programmed Death. Eliminate cells cleanly via intrinsic/extrinsic pathways.

LORE: BAX/BAK cause death (cytochrome c release). BCL-2 prevents it. Venetoclax blocks BCL-2, forcing cancer to die.



Tumour Invasion Ivy

(Invasion and Metastasis)

STAT: EMT (Epithelial-Mesenchymal)



POWER: Metastasis. Breach the capsule to unlock distant colonization.

LORE: Desmoplastic stroma (hard collagen) forms around invasive breast cancer. Lymphovascular invasion drastically upstages prognosis.



E-cadherin

Tumour Architecture Tina

(Tumour Grading)

STAT: Well vs. Poorly Differentiated



Grade 1



Grade 3



POWER: Prognostic Grading. Predict aggressive behavior based on loss of normal structure.

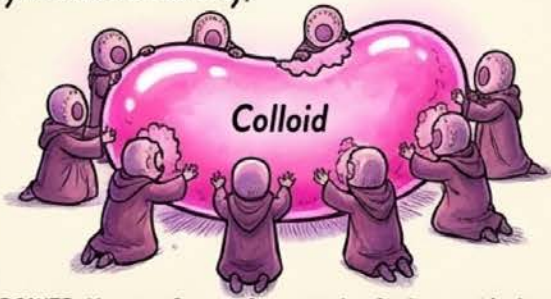
LORE: Breast uses Nottingham (tubules/pleomorphism/mitoses). Prostate uses Gleason (gland architecture).



Thyroid Histology Theo

(Thyroid Microanatomy)

STAT: Extracellular Colloid



Colloid

POWER: Hormone Storage. Store months of iodine outside the cell.

LORE: Follicular adenoma and carcinoma look identical on FNA cytology; you MUST excise the capsule to check for invasion.



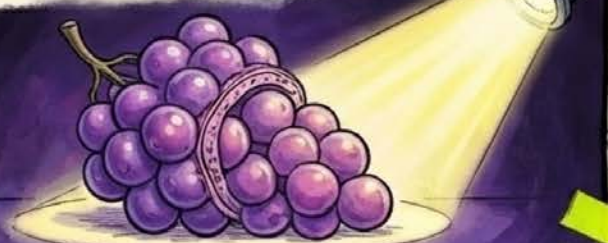
Stay fluffy!



Breast Histology Bella

STAT: *The TDLU*

(Breast Microanatomy)



POWER: Myoepithelial Check. Determine benign vs. invasive status.

LORE: Invasive carcinoma LACKS the myoepithelial layer (p63/SMA negative).
Lobular carcinoma files in single lines because it lost E-cadherin.

The Histopathology Attending

**STAT: The
Ultimate Synthesis**

(Integrated Reasoning)

POWER: Board Exam. Force the player to integrate all previous knowledge.

LORE: Case 1: Lobular Breast Ca. Case 2: Bladder Ca metastatic to node. Case 3: Lupus Nephritis (Full House IF). Integrate or perish!



Stay fluffy!



Prostate Histology Pete

**STAT: Gleason
Grade Groups**

(Prostate Microanatomy)



POWER: Architectural Scoring. Dictate treatment intensity via primary + secondary patterns.

LORE: Gleason 3+4=7 is Grade Group 2 (good). Gleason 4+3=7 is Grade Group 3 (worse). Cribriform (sieve-like) pattern is an automatic Grade 4.

Uncle FrizzleBob's Wildcard

STAT: IHC +
Stains Table

(Synthesis Matrix)

POWER: Cheat Sheet.
When in doubt, apply stain.

LORE: Red is for Collagen (Masson) or Amyloid (Congo). Brown is for IHC. Pink is H&E. Welcome to the Klub, you beautiful degenerates!

[illegible]

KayfaBONGO!
YOU SURVIVED!